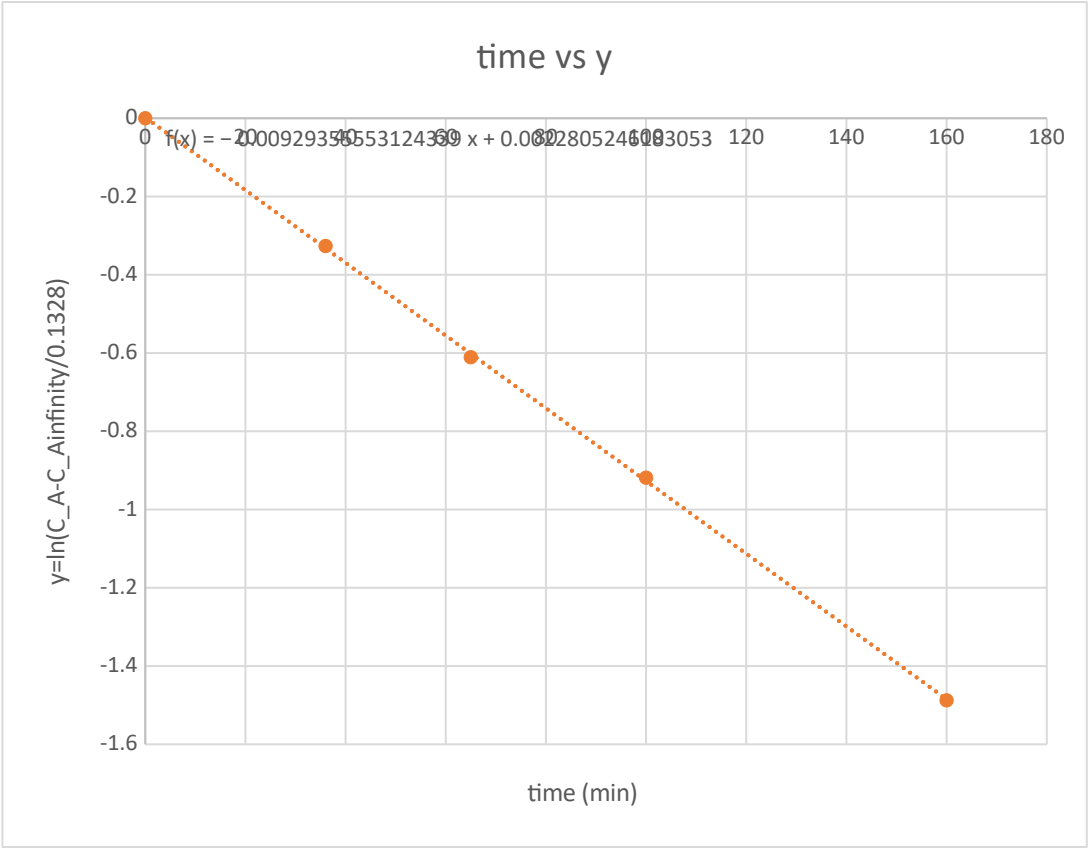


C\_A0 - C\_Ainfinity = 0.1328

| t(min)   | C_A    | C_A-C_Ainfinity | y=ln(C_A-C_Ainfinity/0.1328) |
|----------|--------|-----------------|------------------------------|
| 0        | 0.1823 | 0.1328          | 0                            |
| 36       | 0.1453 | 0.0958          | -0.326581552065518           |
| 65       | 0.1216 | 0.0721          | -0.61079019275143            |
| 100      | 0.1025 | 0.053           | -0.918552323490212           |
| 160      | 0.0795 | 0.03            | -1.48764685538018            |
| infinity | 0.0495 |                 |                              |



average of k = 0.009288



$$\ln(C_A - C_{A\infty} / C_{A0} - C_{A\infty})$$

| k            |
|--------------|
| 0.0090717098 |
| 0.009800298  |
| 0.0087932037 |
| 0.0094849089 |

|                                 | t (min)  | Volume (L) |
|---------------------------------|----------|------------|
|                                 | 120      | 115.455    |
| C <sub>A</sub>                  | 0.093004 |            |
| C <sub>A0</sub> -C <sub>A</sub> | 0.089296 |            |
| mass of B (g)                   | 10.30971 |            |

a) the averages of the k (slope) values check are about the same so it checks off. Slope value is 0.0093min<sup>-1</sup>



b) after 2 hours the tank contains approx 10.3 g of B



29/20

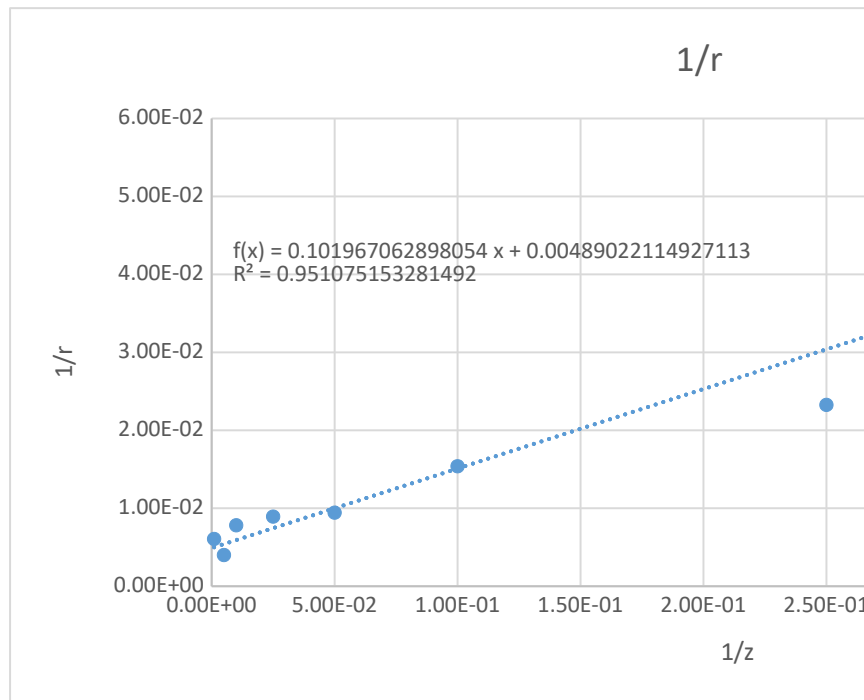
$$= (-)kt \quad .->$$

$$\frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = e^{(-kt)}$$

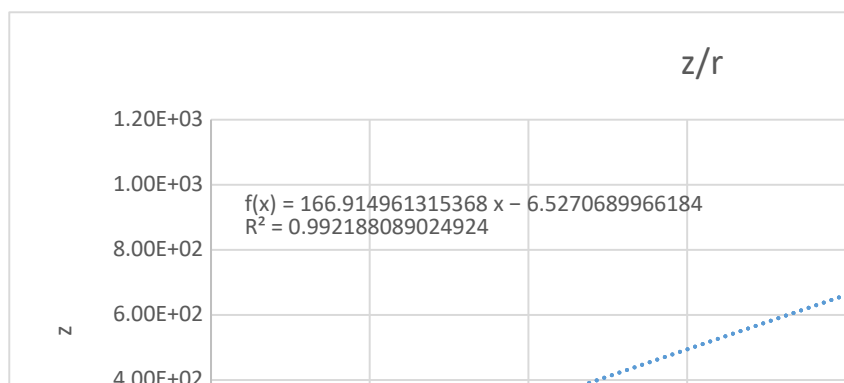
| k      | C <sub>0</sub> -C <sub>∞</sub> |
|--------|--------------------------------|
| 0.0093 | 0.1328                         |

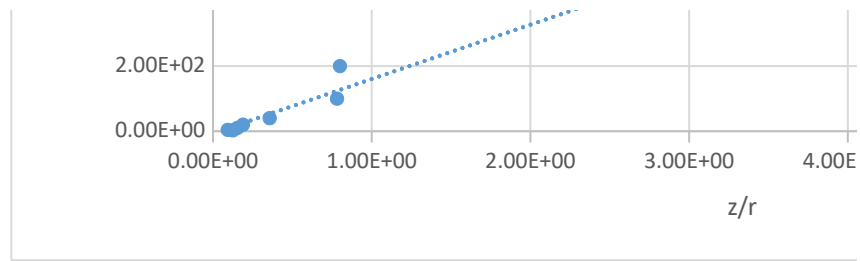
| z        | r   | 1/z      |
|----------|-----|----------|
| 2.50E+00 | 20  | 4.00E-01 |
| 4        | 43  | 2.50E-01 |
| 10       | 65  | 1.00E-01 |
| 20       | 106 | 5.00E-02 |
| 4.00E+01 | 112 | 2.50E-02 |
| 100      | 128 | 1.00E-02 |
| 200      | 250 | 5.00E-03 |
| 1000     | 165 | 1.00E-03 |

1st Process



2nd process





y

B

m

-6.52

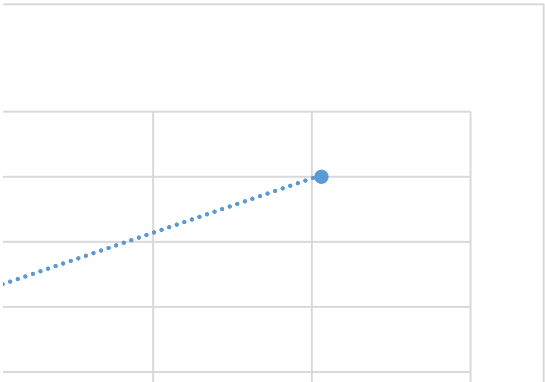
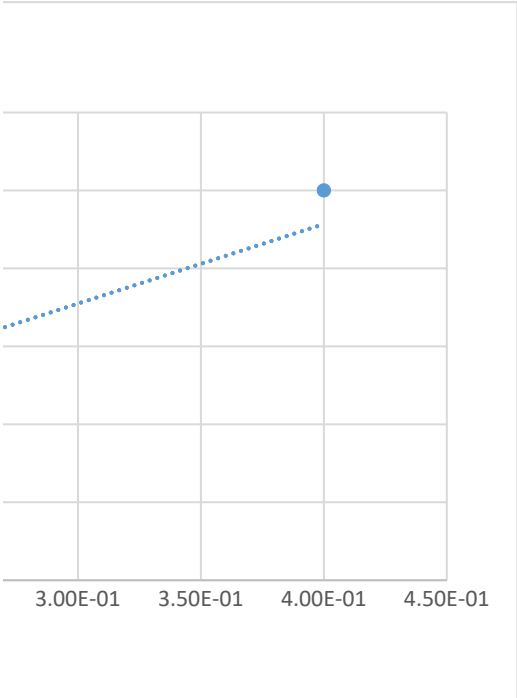
6.52

166.91

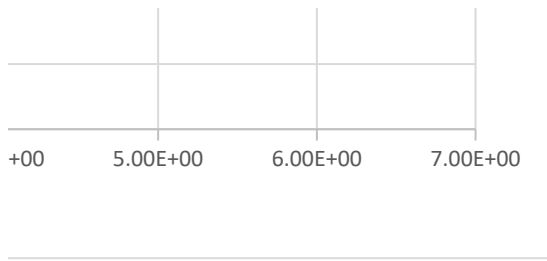
$1/r = (B+z)/(A*z)$   $z=A*(z/r)$

| 1/r      | z/r      |
|----------|----------|
| 5.00E-02 | 1.25E-01 |
| 2.33E-02 | 9.30E-02 |
| 1.54E-02 | 1.54E-01 |
| 9.43E-03 | 1.89E-01 |
| 8.93E-03 | 3.57E-01 |
| 7.81E-03 | 7.81E-01 |
| 4.00E-03 | 8.00E-01 |
| 6.06E-03 | 6.06E+00 |

|                  |             |
|------------------|-------------|
| m = B/A          | y-int = 1/A |
| 0.102            | 0.0049      |
| B                | A           |
| 20.8163265306122 | 204.081633  |



2nd process is more accurate when



A

166.91

A  
185.1852

B  
18.55556



25/25

3 Murphy 1.12

glucose  $\rightarrow$  adipic process loses \$5,400/day

benzene  $\rightarrow$  adipic process makes \$27,100

$$27,100 + 5400 = \$32,500$$

$48,500 - 32,500 = \$16,000$   $\leftarrow$  cost of glucose should drop to

$$\frac{80,850 \text{ kg}}{\text{day}} \cdot \frac{\text{day}}{\$16,000} = \$0.198/\text{kg}$$

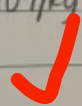


glucose  $\rightarrow$  catechol process makes \$49,200/day

benzene  $\rightarrow$  catechol process nets \$89,300

$$89,300 - 49,200 = 40,100$$

$$\frac{385,579 \text{ kg}}{\text{day}} \cdot \frac{\text{day}}{\$40,100} = \$0.104/\text{kg}$$



$\leftarrow$  glucose price should drop to, to be competitive with benzene

20/20

[4] Murphy 2.11 (1 ton = 2000 lbm)

Hydrocarbon emission was about 14 g/mi      1970     $1100 \times 10^9$  mi  
went to 1 g/mi      2000     $2600 \times 10^9$  mi

Calculate:

→ total hydrocarbon emissions (in tons) for 1970 and 2000

→ Percent increase/decrease over 30 yr period.

$$\frac{14 \text{ g}}{\text{mi}} \times \frac{1100 \times 10^9 \text{ mi}}{1.54 \times 10^{13}} = \frac{1.54 \times 10^{13} \text{ g}}{1.54 \times 10^{13} \text{ g}} \times \frac{1 \text{ ton}}{2000 \text{ lbm}} = 16,975,594.19 \text{ tons}$$

$$\frac{1 \text{ g}}{\text{mi}} \times \frac{2600 \times 10^9 \text{ mi}}{2.6 \times 10^{12}} = \frac{2.6 \times 10^{12} \text{ g}}{2.6 \times 10^{12} \text{ g}} \times \frac{1 \text{ ton}}{2000 \text{ lbm}} = 2,866,009.41 \text{ tons}$$

$$\text{change in emission} = (2.6 \times 10^{12} \text{ g}) - (1.54 \times 10^{13} \text{ g}) = -1.28 \times 10^{13} \text{ g} \text{ or } -14,089,584.7 \text{ tons}$$

percent change in emission

$$\frac{-1.28 \times 10^{13}}{1.54 \times 10^{13}} = -0.83 \times 100 = \sim 83\% \text{ decrease in emission}$$

15/15

$$\rho_{\text{water}} = 0.998 \text{ g/mL} \quad \rho_{\text{ethanol}} = 0.7893 \text{ g/mL} \quad V = \frac{m}{\rho}$$

$$n = 150 \text{ mol}$$

$$\begin{aligned} V_{\text{water}} &= \frac{60 \text{ g}}{0.9982 \text{ g/mL}} = 60.1081 \text{ mL} \\ V_{\text{ethanol}} &= \frac{40 \text{ g}}{0.7893 \text{ g/mL}} = 50.6778 \text{ mL} \end{aligned} \quad \left. \vphantom{\begin{aligned} V_{\text{water}} &= \frac{60 \text{ g}}{0.9982 \text{ g/mL}} \\ V_{\text{ethanol}} &= \frac{40 \text{ g}}{0.7893 \text{ g/mL}} \end{aligned}} \right\} V_{\text{total}} = 110.79 \text{ mL}$$

$$\rho_{\text{mixture}} = \frac{100 \text{ g}}{110.79 \text{ mL}} = 0.902 \text{ g/mL} \quad \text{SG} = \frac{\rho_{\text{mixture}}}{\rho_{\text{water}}} = \frac{0.902 \text{ g/mL}}{0.998 \text{ g/mL}} = 0.904$$

$$M_{\text{ethanol}} = 46.07$$

$$\rightarrow 150 \text{ mol} \times \frac{46.07 \text{ g}}{\text{mol}} = 6,910.5 \text{ g} \quad \frac{6,910.5 \text{ g}}{0.4} = 17,276.25 \text{ g}$$

$$\textcircled{a} \quad V_{\text{mixture}} = \frac{17,276.25 \text{ g}}{\rho_{\text{mixture}}} \rightarrow \frac{17,276.25 \text{ g}}{0.902 \text{ g/mL}} = 19,153.27 \text{ mL or } 19.15 \text{ L}$$

$$\text{SG} = 0.93518 \quad \text{SG} = \frac{\rho_{\text{mixture}}}{\rho_{\text{water}}} \rightarrow 0.93518 = \frac{\rho_{\text{mixture}}}{0.998} \quad \rho_{\text{mixture}} = 0.933 \text{ g/mL}$$

$$\textcircled{b} \quad V_{\text{actual}} = \frac{m}{\rho_{\text{actual}}} \rightarrow \frac{17,276.25 \text{ g}}{0.933 \text{ g/mL}} = 18,516.88 \text{ mL or } 18.52 \text{ L}$$

$$\text{Error} = \frac{V_{\text{assumed}} - V_{\text{actual}}}{V_{\text{actual}}} \times 100 = \frac{19.15 - 18.52}{18.52} = 0.034 \times 100$$

$$\sim 3.4\%$$

25/20